

Arduino IDE Installation Guide

Arduino IDE Windows Installation Guide

Objective

At the end of this guide, the user should be able to:

- Download and install the Arduino IDE
 - Launch the Arduino IDE successfully
 - Configure basic board and port settings
 - Upload a test program to verify proper installation
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System Requirements

Before installing Arduino IDE, make sure your computer meets the following requirements:

Minimum Hardware Requirements

- Processor: Dual-core or higher
- RAM: 2 GB (4 GB recommended)
- Storage: At least 500 MB free space
- USB Port (for Arduino board connection)

Supported Operating Systems

- Windows 10 / 11 (64-bit recommended)
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Required Materials

- Computer (Laptop or Desktop)
 - ESP32 Board
 - USB Cable (Type-A to Type-B or Type-A to Micro-USB / USB-C depending on board)
 - Internet connection
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Step 1 — Download the Arduino IDE

1. Open a web browser.

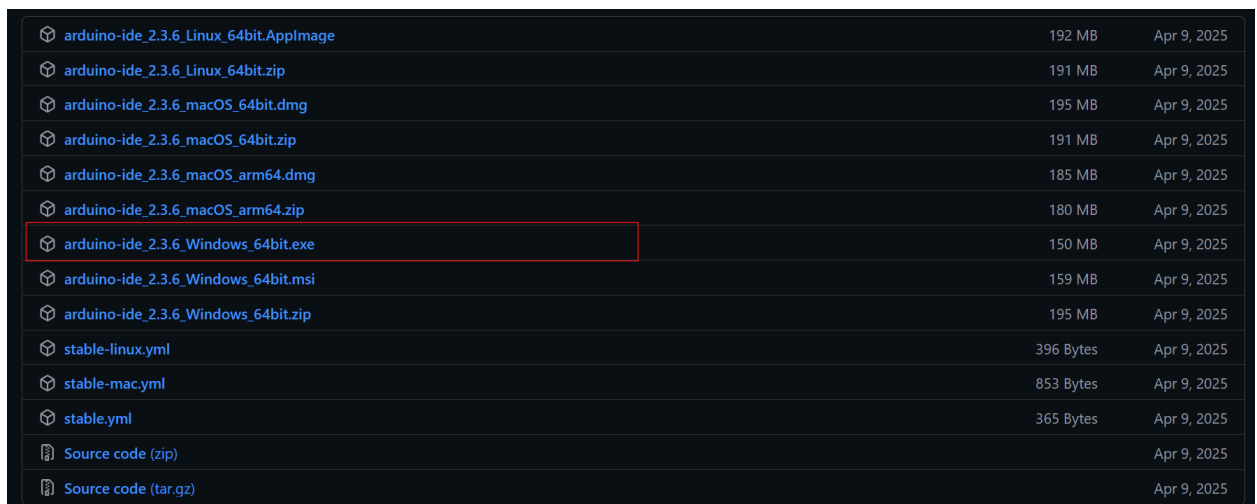
Go to the official Arduino Github website:

<https://github.com/arduino/arduino-ide/releases/tag/2.3.6>

2. Choose the appropriate version of .exe for your operating system:

- **Windows Installer**

3. Click “arduino-ide_2.3.6_Windows64bit.exe”.

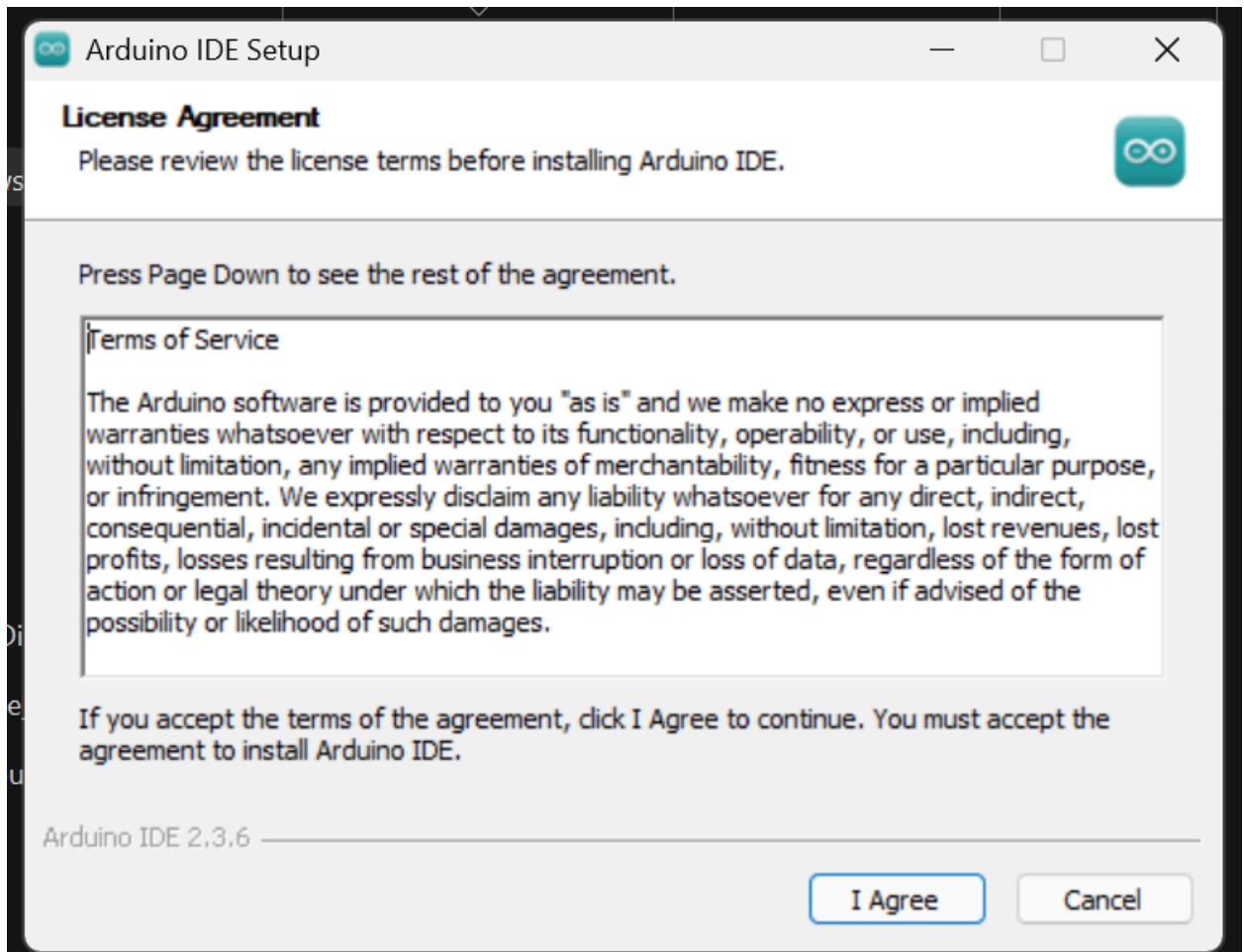


arduino-ide_2.3.6_Linux_64bit.AppImage	192 MB	Apr 9, 2025
arduino-ide_2.3.6_Linux_64bit.zip	191 MB	Apr 9, 2025
arduino-ide_2.3.6_macOS_64bit.dmg	195 MB	Apr 9, 2025
arduino-ide_2.3.6_macOS_64bit.zip	191 MB	Apr 9, 2025
arduino-ide_2.3.6_macOS_arm64.dmg	185 MB	Apr 9, 2025
arduino-ide_2.3.6_macOS_arm64.zip	180 MB	Apr 9, 2025
arduino-ide_2.3.6_Windows_64bit.exe	150 MB	Apr 9, 2025
arduino-ide_2.3.6_Windows_64bit.msi	159 MB	Apr 9, 2025
arduino-ide_2.3.6_Windows_64bit.zip	195 MB	Apr 9, 2025
stable-linux.yml	396 Bytes	Apr 9, 2025
stable-mac.yml	853 Bytes	Apr 9, 2025
stable.yml	365 Bytes	Apr 9, 2025
Source code (zip)		Apr 9, 2025
Source code (tar.gz)		Apr 9, 2025

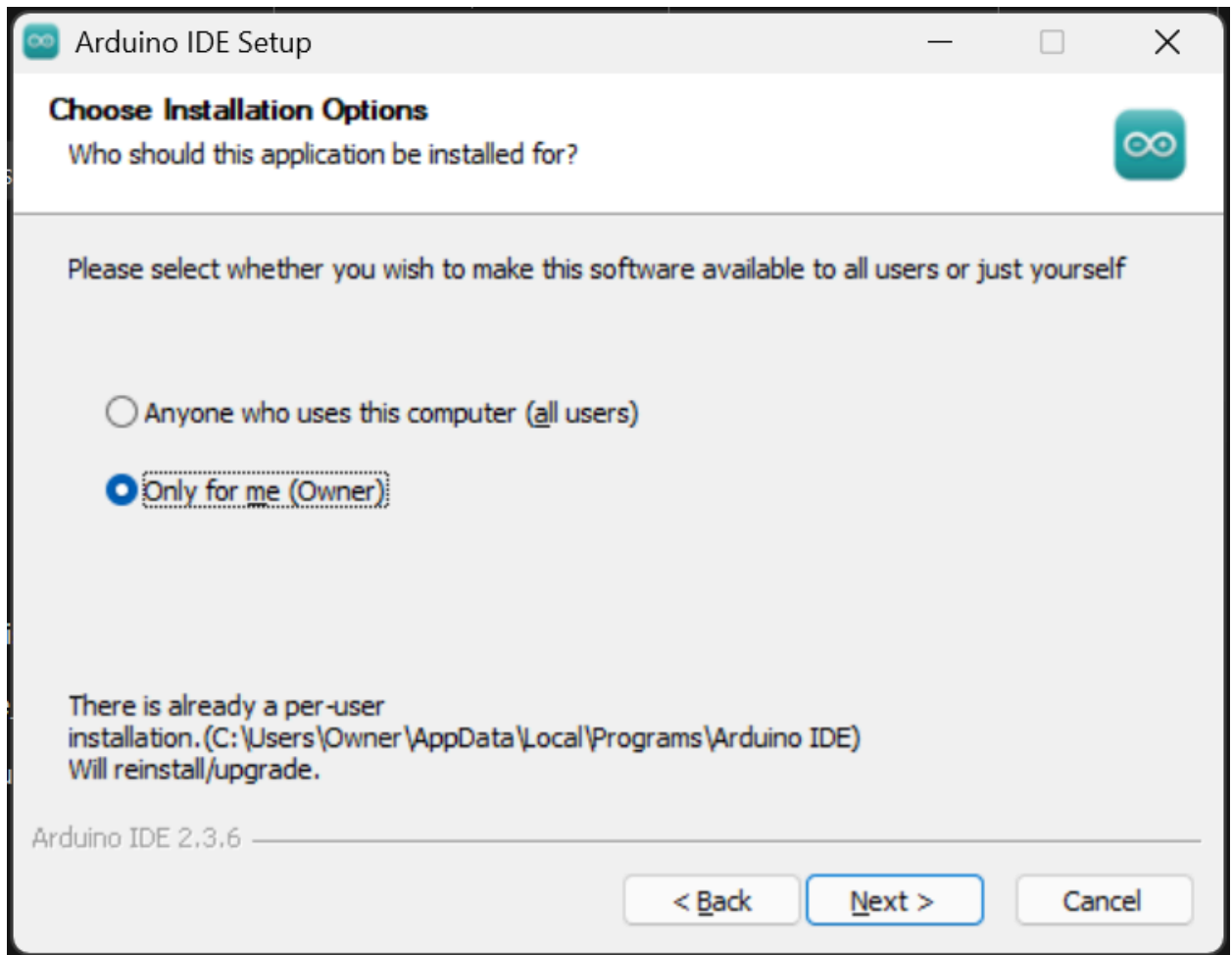
4. If prompted to donate, you may click **Just Download**.

Step 2 — Install Arduino IDE (Windows)

1. Locate the downloaded file (e.g., `arduino-ide_x.x.x_Windows_64bit.exe`).
2. Double-click the installer.
3. Click **I Agree** to accept the license agreement.

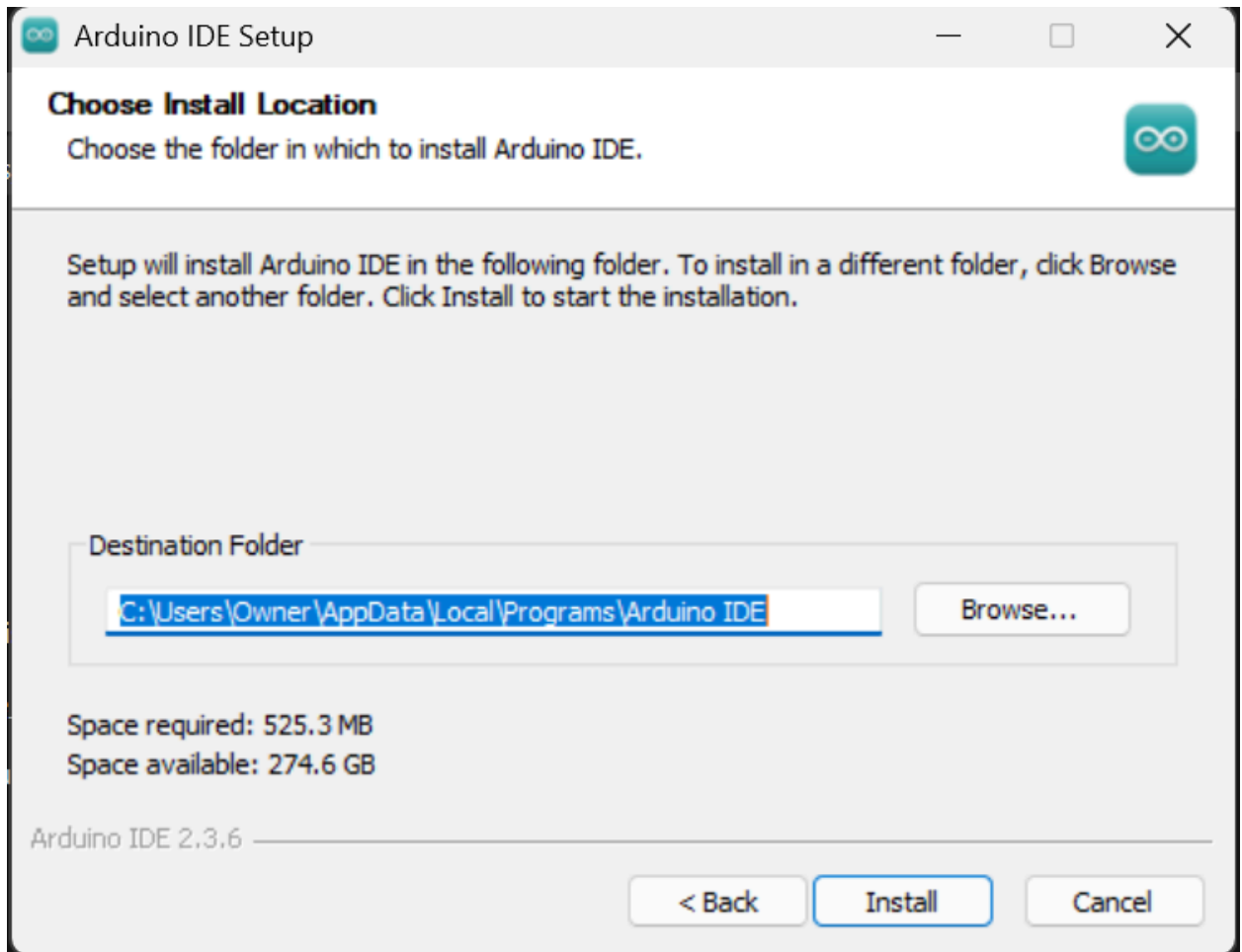


4. Select installation options:



- Anyone who uses this computer for all users
- Only for me, if you have multiple users on your computer.

5. Choose the installation folder (default is recommended).

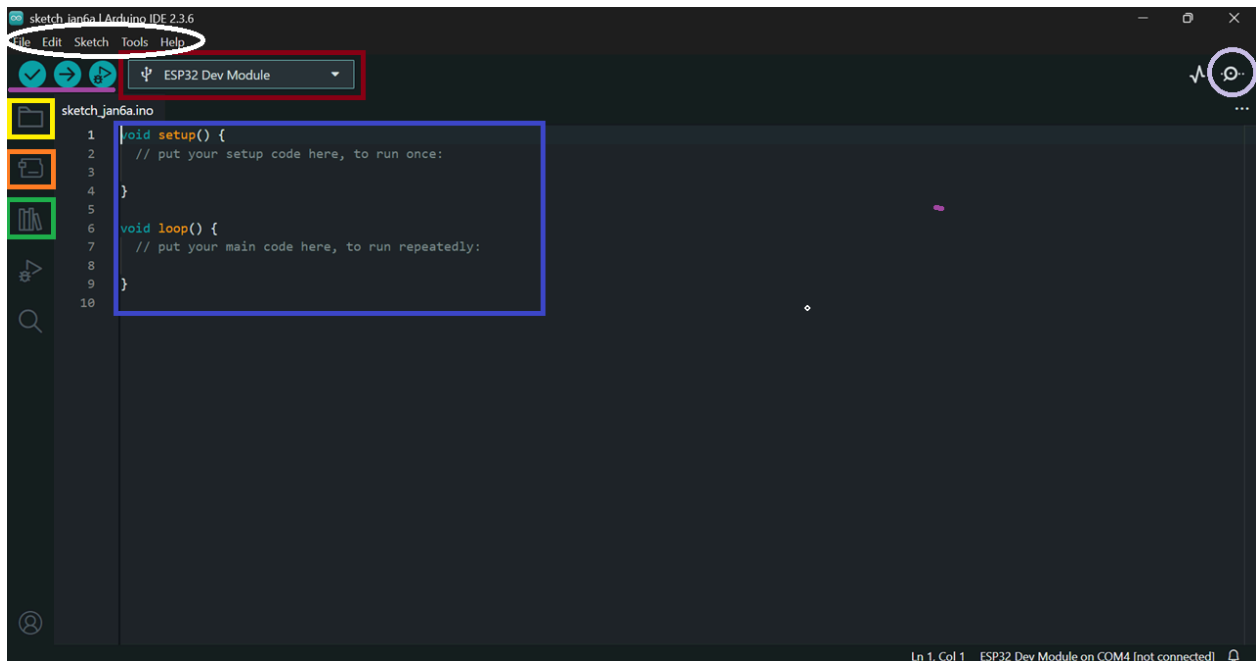


6. Click **Install**.
7. Wait for the installation to complete.
8. Click **Close**.

Step 3 — Launch the Arduino IDE

1. Open **Arduino IDE**.

2. The main window should display:



Arduino IDE Legend:

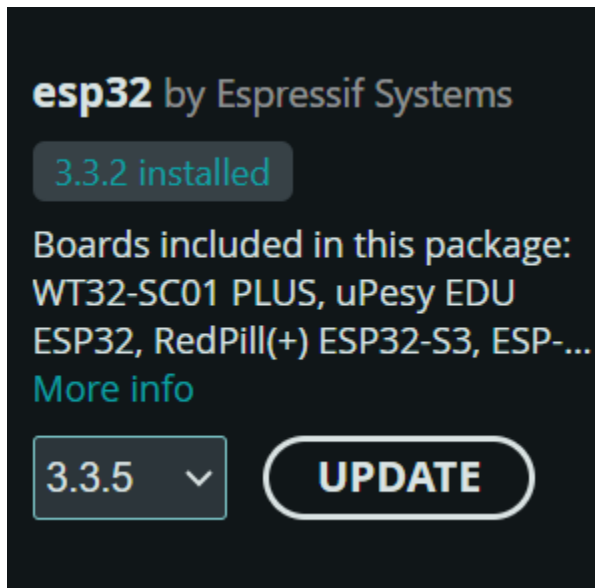
- Indigo Box = Code Editor
- White Circle = Toolbar (File,Edit,Sketch,Tools,Help)
- Purple Line = (As follows Verify,Upload,Start Debugging)
- Lavender Circle = Serial Monitor
- Yellow Box = Sketchbook
- Orange Box = Boards Manager
- Green Box = Library Manager

✓ If the IDE opens without errors, installation is successful.

Step 4 — Install ESP32 Board

NOTE: Ensure that CH340 Driver is installed on the PC/Laptop to be able to detect the ESP32

1. Click the board manager , Type “esp32” ,select and install the 3.3.2 version:



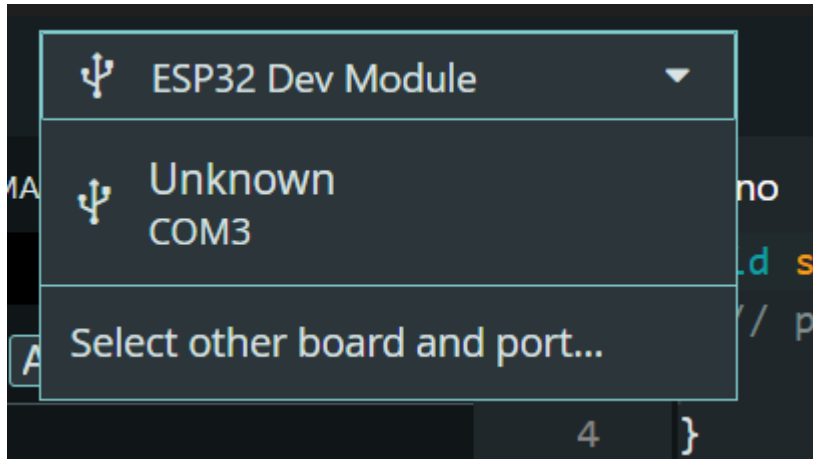
2. Ensure the installed version is “3.3.2”
 3. Connect the trainer board , determine its com port(which varies per PC or laptop) and select ESP 32.
 4. Once the trainer board is properly connected , try to upload
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Step 5 — Select the Board and the PORT

In Arduino IDE, click:

Tools → Board

1. Choose your board:
 - ESP32 Dev Module (must be installed)



2. If your board is not listed, additional board packages may be required.
3. Select the COM port: Windows: COM3, COM4, etc.
4. If no port appears, check: USB cable, Driver installation & Board connection

Step 6 — Upload a Test Program (Blink)

Open:

File → Examples → 01.Basics → Blink

1. Click **Verify** (✓) to compile the code.
2. Click **Upload** (➡) to upload the program.
3. Observe the onboard LED blinking every second.

✓ Blinking LED confirms successful installation and configuration.

Common Troubleshooting

Problem	Possible Solution
Port not showing	Install drivers, try another USB cable
Upload error	Check board & port selection
Board not detected	Restart IDE, reconnect board

Safety and Best Practices

- Always disconnect the board before rewiring
- Use proper USB cables
- Avoid powering external components directly from the board without checking current limits

Library Installation Guide

Arduino IDE Library Installation Guide

Arduino libraries add extra functionality to your projects (for sensors, displays, WiFi, LoRa, RFID, etc.). Installing Libraries are essential to be able to upload the code.

Install Using Library Manager (Recommended)

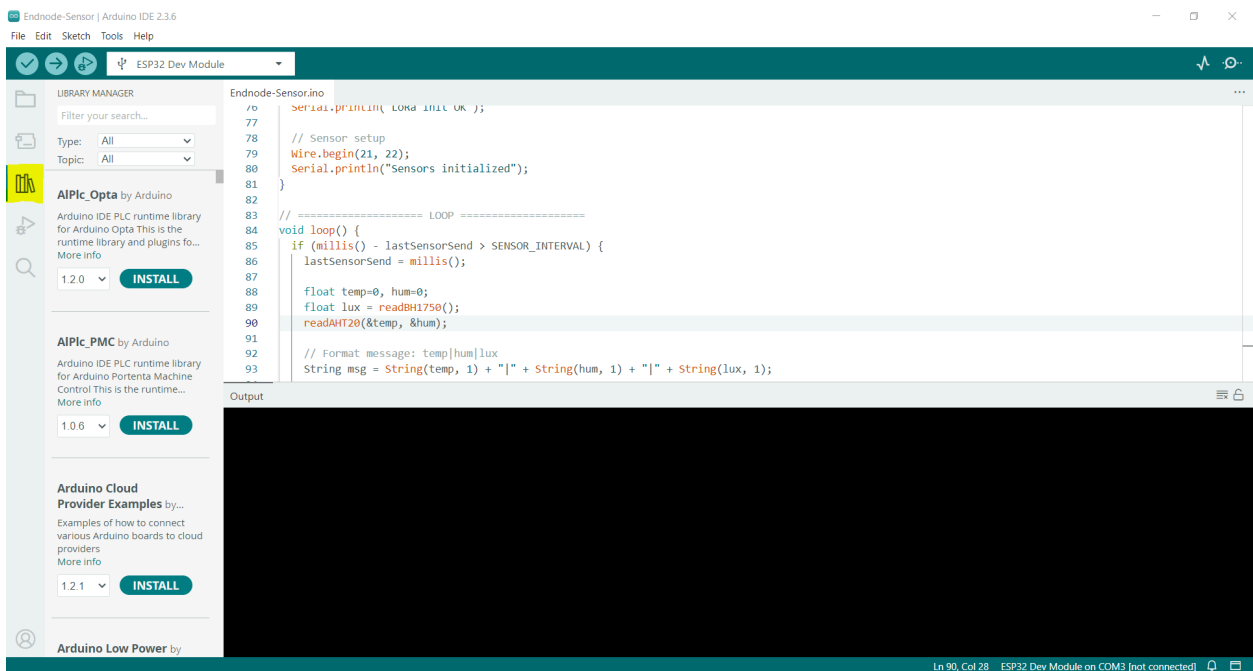
This is the easiest and safest method.

Steps:

1. Open Arduino IDE

Click:

Library Manager Icon



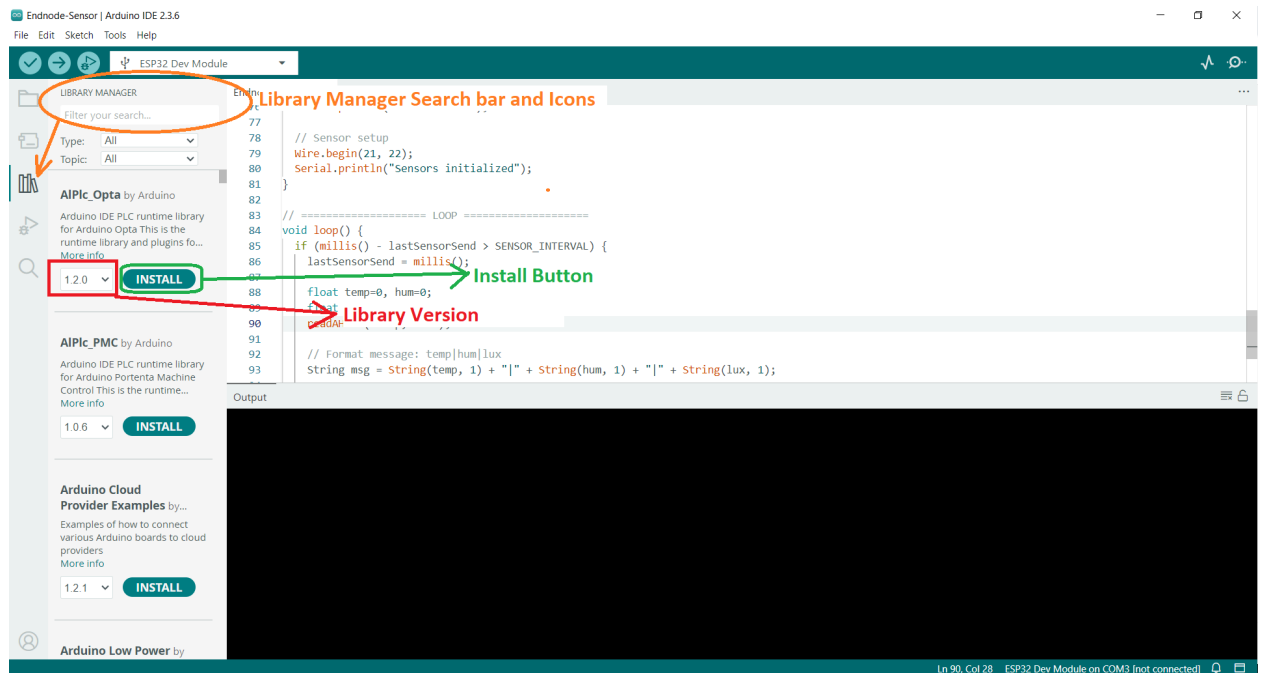
2. The **Library Manager** window will open.

In the search bar (top right), type the library name

Example Libraries:

LoRa.h

3. Select the correct library and correct version from the list



4. Click **Install**

Wait until it shows:

Installed

5. Restart Arduino IDE (recommended).

Note before you start your Trainer Journey Install this libraries:

PxMatrix-1R by Aleksandra Menschikova V1.2.0

DFRobotDFPlayerMini by DFRobot V1.0.6

NOTE: Install CH340 Drivers if you don't have it yet! [CH340 Link](#)